PYTHON PROGRAMMING FOR BEGINNERS
A Clear, Simple, and Comprehensive Guide to Python Fundamentals

1. What is Python?
Python is a high-level, interpreted programming language created by Guido van Rossum and released in 1991. Its design philosophy emphasizes code readability with the use of significant indentation.
- Easy to Learn: Simple syntax that is great for beginners.
- Versatile: Used in web development, data science, automation, AI, and scripting.
- Huge Community: Thousands of free libraries and active community support.

2. Variables and Data Types
A variable is a container for storing data values. You create a variable the moment you first assign a value to it.
```
# String (Text)
name = "Alice"

# Integer (Whole Number)
age = 25

# Float (Decimal Number)
price = 19.99

# Boolean (True or False)
is_active = True
```

3. Basic Math Operations
Python can be used as a standard calculator to perform arithmetic operations.
```
x = 10
y = 3
print(x + y)  # Addition: 13
print(x - y)  # Subtraction: 7
print(x * y)  # Multiplication: 30
print(x / y)  # Division: 3.333...
print(x % y)  # Modulus (Remainder): 1
```

4. Input and Output
Interacting with users requires displaying output and capturing input.
```
# Output data to screen
print("Hello, World!")

# Take input from user
username = input("Enter your name: ")
print("Welcome, " + username)
```

5. Conditional Statements (If-Else)
Conditional statements let your program make decisions based on whether a condition is true or false.
```
score = 82
if score >= 90:
    print("Grade: A")
elif score >= 75:
    print("Grade: B")
else:
    print("Grade: C")
```

6. Loops
Loops repeat a block of code multiple times. Python provides for loops and while loops.
```
# For loop
for i in range(3):
    print("Iteration count:", i)
```

```
# While loop
count = 2
while count > 0:
    print("Countdown:", count)
    count -= 1
```

7. Functions
A function is a reusable block of code designed to perform a specific task when called.

```
def greet_user(name):
    return "Hello, " + name + "!"

message = greet_user("Bob")
print(message)
```

8. Lists
Lists store multiple items in a single variable. Items are ordered, changeable, and allow duplicate values.

```
fruits = ["apple", "banana", "cherry"]
fruits.append("orange")  # Add item
print(fruits[0])          # Access first item (apple)
```

9. Dictionaries
Dictionaries store data values in key-value pairs. Each key must be unique.

```
student = {"name": "John", "age": 22, "major": "Computer Science"}
print(student["name"])    # Output: John
```